

Design and Analysis of Microstrip Patch Antenna for 2.4GHz ISM Band and WLAN Application

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Abstract— In this paper Microstrip Patch antenna is designed in Ansoft HFSSV13. The designed antennas are double-sided printed microstrip patch antenna. The antenna is designed over the operating frequency is 2.4GHZ using the substrate material as FR4 which has the dielectric constant of 4.4(ϵ_r). The designed antenna can be used for ISM (industrial, scientific and medical) band and WLAN applications. The designed antenna has low profile, low cost, easy fabrication and good isolation. The antenna is designed by using Ansoft HFSSV13 simulation software and designed antenna provides return loss less then -10dB. The parameters such as return loss, voltage standing wave ratio (VSWR), mutual coupling, gain, radiation pattern has been simulated and analyzed.

Index Terms— Industrial Scientific and Medical (ISM), HFSS (High Frequency Structural simulator), Double-sided printed.

I. INTRODUCTION

Microstrip patch antenna is a key building in wireless communication and Global positioning system. A Patch antenna is a radio antenna with low profile, which can be mounted on a flat surface [1]. A patch antenna is a narrowband and it consists of a flat rectangular sheet mounted over a larger sheet of metal called as ground plane.

Patch antenna is simple to fabricate and easy to modify and customize. Compared with conventional antenna microstrip patch antenna have advantages [1] [2]. They are light weight, low cost and low profile planer configuration.

A microstrip patch antenna is well suited for wireless communication [9] due to its light weight and low planer configuration. The impedance bandwidth [7] [11] of a patch antenna is the spacing between the patch and ground plane. Patch arrays can provides higher gain then a single patch. Such an array of patch antenna is an easy way to make a phases array of antenna with dynamic beamforming ability.

The industrial, scientific and medical (ISM) bands are operated at the frequency range 2.4GHZ for industrial, scientific and medical purposes. The antenna is designed for 2.4 GHZ and it can be used for various applications like WLAN, Wi-Fi family of standard (802.11 a, b, g, n) and Bluetooth short range wireless [8] application.

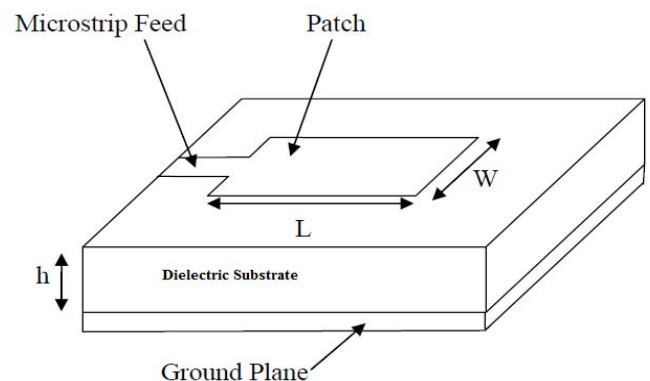


Figure 1. Microstrip patch antenna

The most commonly employed microstrip patch antenna is rectangular patch antenna. Patch antenna is the ability to have polarization diversity. If the dielectric constant of the substrate increases, the antenna beam width decreases.

In section II, the single antenna is designed. In section III, single antenna designed results and single antenna design structure with Increasing thickness of the substrate are shown in Fig 2&8 are discussed. In section IV, discuss about conclusion. At last detail about reference.

II. SINGLE ANTENNA DESIGN

The designed antenna is printed on both sides, one side is patch and other side is ground plane. Microstrip patch

antennas can be fed by a variety of methods. The antenna is designed by using a microstrip feed line because it is one of the easier method to fabricate. The designed antenna is used as a substrate material is FR-4. The dielectric constant of the substrate material is 4.4. The geometry of the given antenna is illustrated in Fig.2.

The antenna is fabricated on 76.8 X 57.8 mm² FR-4 substrate with a dielectric constant of 4.4 (ϵ_r) and substrate thickness 1.6mm.

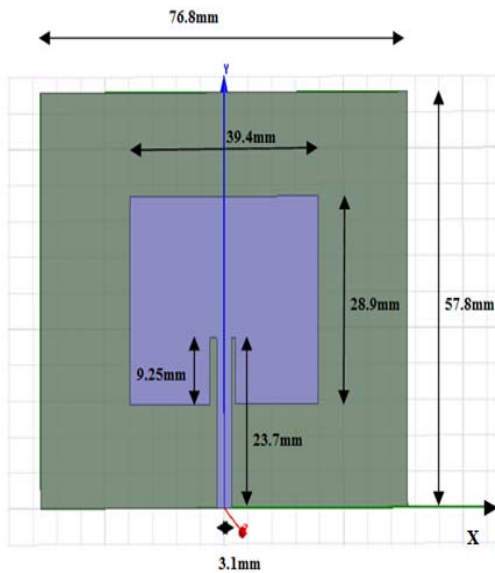


Figure2. Geometry of antenna with dimensions in mm (front view)

The patch has dimension on 39.4X28.9mm² and it is fed by a microstrip line having width of 3.1mm.

III. SIMULATION RESULTS AND DISCUSSION

A. Single Antenna Design

In this section, the antenna parameters such as return loss, gain, voltage standing wave ratio (VSWR), radiation pattern, directivity and mutual coupling.

(1) Return Loss

Return loss is the loss of power in the signal reflected or returned by the discontinuity of transmission line. It is related to the reflection coefficient and standing wave ratio (SWR). Return loss can be expressed as,

$$RL = -20 \log [\Gamma]$$

Where,

Γ - Reflection Coefficient Here designed antenna provides return loss at 2.4GHZ is -28.5dB. Return loss at 4.5GHZ is -24dB and -22.5dB at 6.1GHZ.

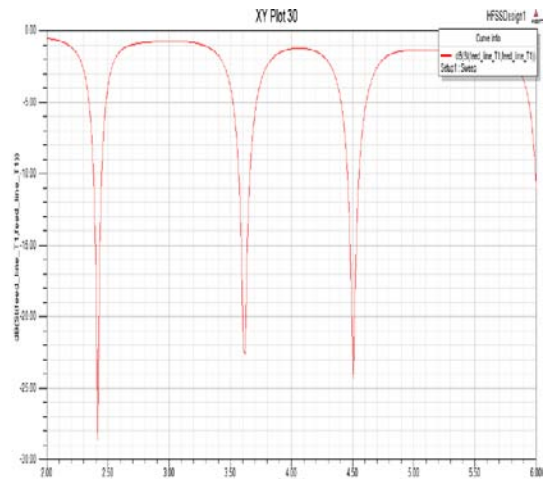


Figure 3. Return loss for Microstrip patch antenna

(2) Bandwidth

The range of Bandwidth describes frequencies over which the antenna can properly radiate or receive energy.

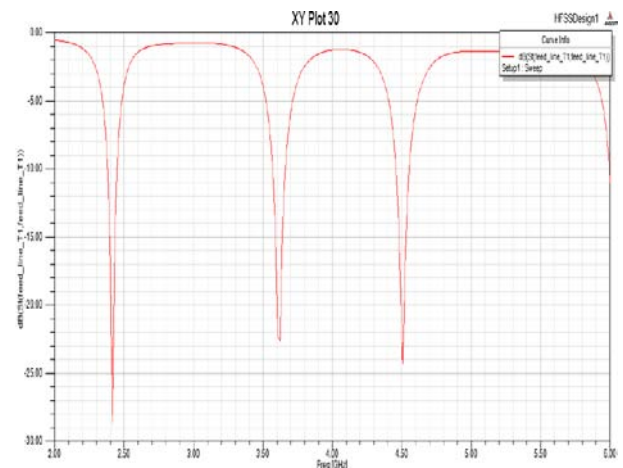


Figure 4. Bandwidth for Microstrip patch antenna

Here designed antenna provides the bandwidth 80 MHz at 2.4 GHZ, 110 MHz at 3.6 GHZ, 120 MHz at 4.5 GHZ and 130MHz at 6.1GHZ.

(3) Gain

Gain is the ratio of radiation field intensity of test antenna to that of the reference antenna. The gain of the designed antenna is 3.97 dB.

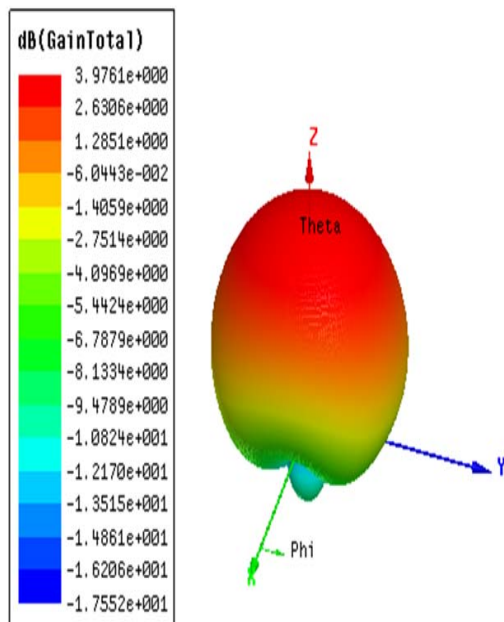


Figure 5. Gain for Microstrip patch antenna

(4) Voltage Standing Wave Ratio

It is defined as a measurement of the mismatch between the load and the transmission line. For ideal case the value of VSWR is 1 and for better matching.

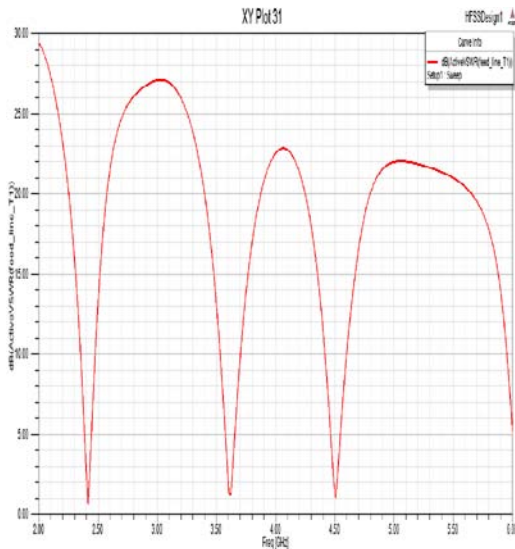


Figure 6. VSWR for Microstrip patch antenna

Voltage standing wave ratio of this antenna is 1.6 at 2.4GHz, 1.3 at 6.1GHz.

(5) Radiation Pattern

Radiation patterns are the graphical representation of the electromagnetic power distribution in free space.

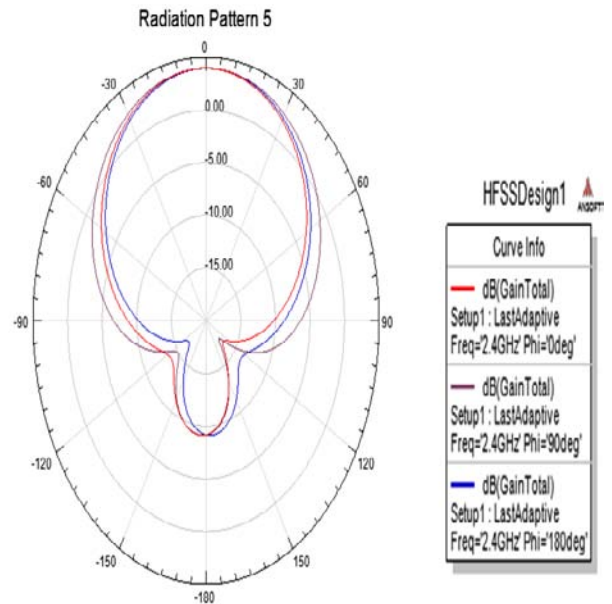


Figure 7. Radiation pattern for Microstrip patch antenna

B. Single Antenna Design with Increasing Thickness of the Substrate

Here the single antenna is designed with increasing thickness of the substrate and thickness of the substrate is 1.8mm.

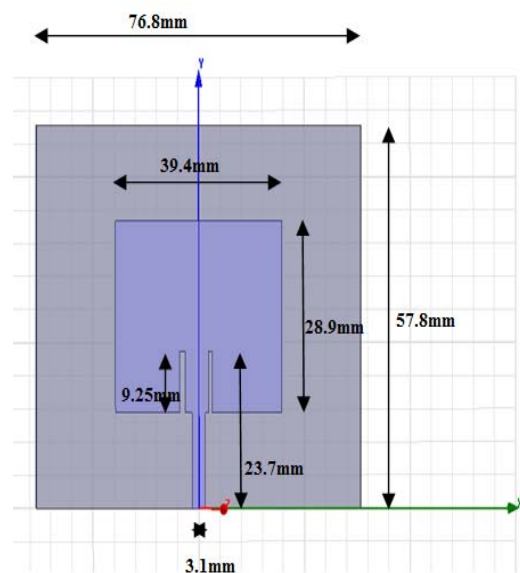


Figure 8. Geometry of antenna with dimensions in mm (front view)

C. Simulation Analysis

In this section, the antenna parameters such as return loss, gain, voltage standing wave ratio (VSWR), radiation pattern, directivity and mutual coupling.

(1) Return Loss

It is a measure of the effectiveness of an antenna to deliver of power from source to the antenna. For various resonant frequencies with bandwidth is obtained.

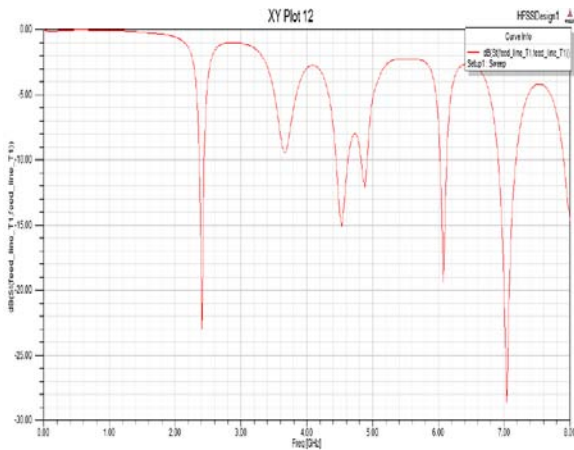


Figure 9. Return loss for Microstrip patch antenna

4.5GHZ, -12dB at 4.9GHZ and -19dB at 6.1GHZ.

(2) Bandwidth

Here designed antenna provides the bandwidth 90 MHZ at 2.4 GHZ, 210 MHZ at 4.5 GHZ, 100 MHZ at 4.9 GHZ and 160MHZ at 6.1GHZ.

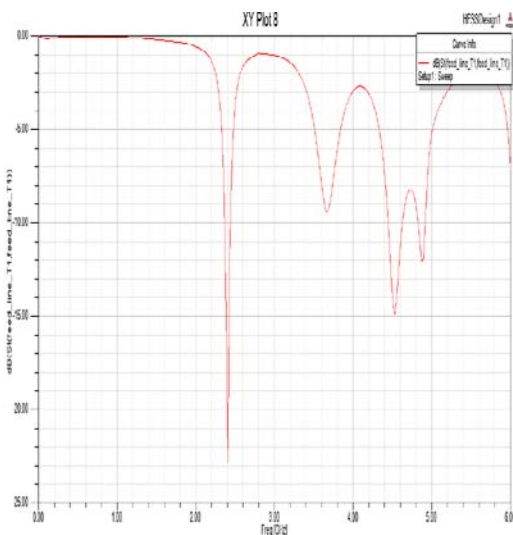


Figure 10. Bandwidth for Microstrip patch antenna

(3) Gain

Gain as a parameter measures the directionality of antenna. Here gain of the antenna is 4.22 dB.

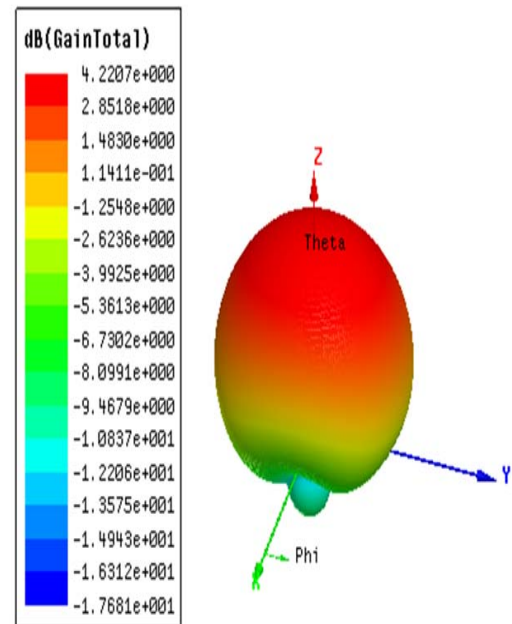


Figure 11. Gain for Microstrip patch antenna

(4) Voltage Standing Wave Ratio (VSWR)

It is defined as a measurement of the mismatch between the load and the transmission line. For ideal case value of VSWR is 1 and for better matching. The VSWR of designed antenna is 1.2 at 2.4GHz and 1.8 at 6.1GHz.

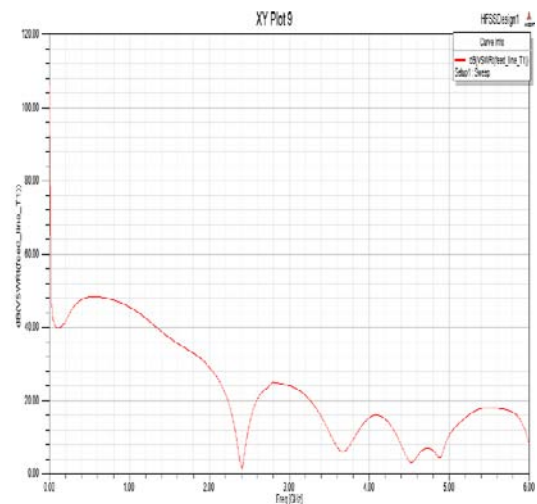


Figure 12. VSWR for Microstrip patch antenna

(5) Directivity

Directivity is a figure of merit for an antenna. Directivity measures power density of an antenna that radiates in the direction of strongest emission.

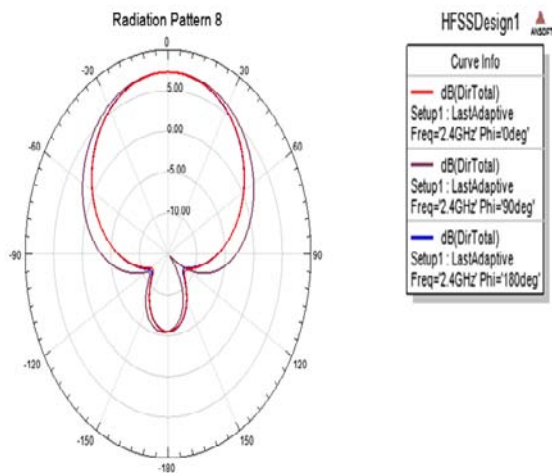


Figure13. Directivity for Microstrip patch antenna

The directivity of the proposed antenna is 7.22dB at phi equals to 0, 90 & 180 degree at 2.4GHZ.

D. Applications\

The designed antenna is used for ISM (industrial, scientific, medical) and WLAN.

IV. CONCLUSION

Finally Microstrip patch antenna is designed for 2.4GHz using HFSS simulation software. Two configuration of antenna is proposed. The first configuration is single antenna design structure. The second configuration is single antenna design structure with increasing thickness of the substrate. The designed antenna structure operated at various frequency bands like 2.4 GHz, 3.6 GHz, 4.5 GHz, 6.1GHZ and 7.1GHZ. Those frequency bands can be applied for various applications. The applications are ISM band, WLAN[8]. The return loss has been achieved -23dB at 2.4 GHZ. Further the antenna is designed using antenna array [12] [13].

TABLE.1 COMPARISON BETWEEN EXISTING METHOD AND PROPOSED METHOD

S.NO	PARAMETER	EXISTING METHOD	PROPOSED METHOD
1.	GAIN	3.97dB	4.22dB

2.	VSWR	2.4GHZ: 1.6 6.1GHZ: 1.3	2.4GHZ: 1.2 6.1GHZ: 1.8
3.	RETURN LOSS	2.4GHZ: -28.5dB 4.5GHZ: -24dB 6.1GHZ: -22.5dB	2.4GHZ: -23dB 4.5GHZ: -15dB 6.1GHZ: -19dB

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